Federico Palatnik Moroz

Backend & Integrations Engineer · Forward Deployed Engineer · AI Integration — I own the whole service and leave it ready for someone else to maintain

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· github.com/federicomoroz · federicomoroz.github.io · case studies · Remote · UTC-3, full overlap with US business hours

PROFESSIONAL SUMMARY

I build backend services and integrations for systems other people run every day. At an international e-commerce operation I replaced a ~10-minute manual process with one that takes **5 seconds across ~2,100 monthly transactions**, and I kept the platform behind it running —11 tables, ~115,000 records and 48 production automations— alongside the team that used it daily.

I work across **Python, C#/.NET and Java**: the problem picks the stack, not the other way round. I write the code the way a team's code gets written —ADRs recording what was rejected, a runbook, the protocol spec— and I measure rather than estimate. **The code I deliver to clients is private**; the systems linked below are the ones I can show with the code open.

TECHNICAL SKILLS

Languages: Python · C# 12 · Java 21 · TypeScript · JavaScript · SQL

Backend and APIs: FastAPI · Flask · SQLAlchemy · Pydantic · ASP.NET Core (MVC, Web API and minimal APIs) · Entity Framework Core (migrations, `AsNoTracking`) · Spring Boot · Spring Kafka · Node.js · React · REST · OpenAPI/Swagger · WebSockets · SSE

Integrations and automation: Amazon · Mercado Libre · Tienda Nube · couriers · tax authority · Airtable (Scripting and Interface Extensions SDK) · webhooks · retries with exponential backoff · idempotency · timeouts and cancellation · secret handling and IP allowlisting

Data and infrastructure: MySQL · PostgreSQL · Redis · SQLite · Apache Kafka · Docker · Docker Compose · GitHub Actions · versioned migrations (EF Core, Flyway) · multi-stage Docker image running as non-root · container deployment (Render) · `live` / `ready` health checks

Architecture: SOLID · Clean Architecture / Hexagonal (ports & adapters) · MVC · Repository · command/query separation · Event-driven · Transactional Outbox · Idempotent consumer · Circuit breaker · Backpressure

Testing and operations: pytest · xUnit · JUnit · ArchUnit · multi-engine contract tests · real MySQL and PostgreSQL in CI via Testcontainers · `WebApplicationFactory` · purpose-built load tests · structured logging · warnings as errors · runbook and ADRs

PROFESSIONAL EXPERIENCE

Backend Developer
2022 – Present

Freelance · Remote

Custom development for clients —backend APIs and services, carrier and marketplace integrations, operational process automation —: I scope the problem with the people who will use the system, build it, and leave it running. **Much of what I deliver stays private**; what follows is what I can show with the code open.

> **Nexo — B2B catalogue synchronisation** .NET 8 · ASP.NET Core MVC · EF Core · MySQL · PostgreSQL · Redis · WebSockets · SSE · OpenID Connect
>
> github.com/federicomoroz/nexo · case study
>
> An ERP publishes catalogue, prices and stock, and a reseller network consumes them over two transports on the same domain: HTTP for point queries and a WebSocket with a purpose-built frame protocol for bulk pulls, with per-batch acknowledgement backpressure. A seven-step authorization pipeline shared by both transports, with HMAC-SHA256 and a quota shared in Redis, and a Razor operations panel over SSE.
>
> Measured, not estimated: pulling 48,000 items costs **1 authorization instead of 96** against the same work over paginated HTTP; and in a separate run against 8,000 items, 40 simultaneous connections all complete the pull while an ERP change reaches every one at **p50 68 ms**, and that latency **does not move between 1 and 40 connections**: the cost is in the write path, not the fan-out. **356 tests** in CI — 60 domain, 93 API, 95 application and 108 contract — with MySQL and PostgreSQL in real containers.

> **Shipping Quote — multi-carrier shipping rate quoter** Python · FastAPI · SQLAlchemy · Alembic · .NET 8 · ASP.NET Core · EF

Core · MySQL · Testcontainers
shipping-quote.onrender.com — live demo · Python repo · .NET repo · case study

Quoting a shipment means asking each carrier separately and waiting for the slowest, and one carrier going down cannot take the whole quote with it. The service queries all three within the same request and returns whatever came back. It exists in two implementations —Python and .NET— over the same ports, use cases and adapters; the live demo animates the eight hops each request makes through the circuit. The carriers are queried in parallel and **a test measures the clock and fails past 700 ms**: in parallel it is 306, sequential would be 900. Async end to end, cancellation threaded down to the adapter, and a middleware mapping domain exceptions to HTTP status codes. **31 tests in Python and 63 in .NET**.

Order Outbox Service Java 21 · Spring Boot · Kafka · PostgreSQL
github.com/federicomoroz/order-outbox-service · case study

An order is saved and another service has to be told. If the broker is down at commit time the event is lost and the order goes unnotified; retry blindly and the notification goes out twice. Solved with a transactional outbox: the order and its event are written in a single Postgres commit and a separate relay publishes them to Kafka with backoff, so a broker outage cannot take the order down with it. The consumer dedupes by `event_id`, turning the broker's at-least-once into exactly-once at the business level. **86 tests** (72 unit + 14 integration) and ten ArchUnit.

AI training plan generator Flask · Claude API · SSE · Docker
federicomoroz.github.io/en/projects/mega-training-system/

An instructor was building every indoor cycling class and strength plan by hand, each with the methodology and exact duration its discipline demands. The platform generates them with the Claude API and today runs in a gym's daily operation. SSE streaming so the plan appears as it is written, a circuit breaker around the model calls, idempotent generation, and a plugin architecture that allows new disciplines without touching the core.

Systems and automation
2010 – Present

Inducor Corrugados · Corrugated cardboard manufacturing

I have written and maintained the systems behind the operation for fifteen years: purchase orders, delivery notes and logistics, a circuit that used to run on paper. **It started on Excel and I have rebuilt it with each new stack I picked up**, with the business running throughout.

Backend Developer / Automation Engineer
Nov 2024 – Mar 2026

Del Mundo a Tus Manos · International e-commerce and logistics · In-house, full-time

I designed and maintained the operations platform of an international e-commerce business on Airtable, working alongside the operations team that used it every day: a data model of 11 linked tables, ~115,000 records and 48 automations, with the operator-facing interfaces built on top in Node.js, React and TypeScript, and integrations with Amazon, Mercado Libre, Tienda Nube, the national tax authority API, international couriers and a legacy Laravel/PHP backend, handling authentication, secrets management and IP allowlisting. **I cut operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.** I diagnosed and fixed a silent failure on batches of 100+ records — requests hung with no trace in the execution log — with an AbortController, retry with exponential backoff and non-blocking per-record error handling.

Game Developer and Programming Instructor
2021 – 2024

Freelance

Systems, components, plugins and tooling for game designers: **C#** on Unity, **C++** and Blueprints on Unreal. One-to-one programming teaching with **code review** on each student's code.

Owner — Musical instrument import and retail
2006 – 2017

Own business · Buenos Aires

I ran the business end to end: overseas suppliers, importing, stock and sales. **I automated the whole administrative side in Excel** —import costing, margins and risk, assets and liabilities, the accounting books— on the accounting, business-administration and foreign-trade coursework from my degree. It is where I learned to read a process from the side of the person using it, as I later did at Del Mundo a Tus Manos.

EDUCATION AND LANGUAGES

Education: Escuela Da Vinci — Programming (2021 – 2023) · UADE — Marketing, School of Economics (2007 – 2010)

Languages: Spanish — Native · English — Full Professional Proficiency (C2)